CE205 Databases and Information Retrieval

Lab (Week 7)

Objective:

The purpose of this activity is:

- **text/document processing**

Instructions:

Text Processing and Training Phase

1. Read the Factory Reports data from the file "factoryReports.csv" using the following statements:

```
filename = "factoryReports.csv";
data = readtable(filename,'TextType','string');
```

The Factory Reports dataset is a table containing 480 reports with various attributes including a plain text description in the variable Description and a categorical label in the variable Category.

*(form the workspace, open and see the contents of variable **data**)*
OR
See the first 8 records using the following command:

```
head(data)
```

2. Extract the text data and the labels from the **Description** and **Category** columns, respectively, using the following statements:

```
textData = data.Description;
labels = data.Category;
```

*(from the workspace, open and see the contents of variable **`textData`** and **`labels`**)*

- To see the first ten documents, use the following statements:

```
textData(1:10)
```

- To analyse the dataset, change the data type of variable ***`labels`*** (from string to categorical) using the following statement:

```
cLabels=categorical(labels);
```

- Now, plot histogram to see the frequency of each type of document in the dataset using the following statements:

```
figure
histogram(cLabels)
xlabel("Document Type")
ylabel("Frequency")
title("Document Distribution")
grid on
```

3. Tokenize the documents using the following statement:

```
cleanedDocuments = tokenizedDocument(textData);
```

- Now, again display the first ten documents using the following statement:

```
cleanedDocuments(1:10)
```

4. Remove stop words words like "a", "and", "to", and "the" using the following statement:

```
cleanedDocuments = removeStopWords(cleanedDocuments);
```

- Now, again display the first ten documents using the following statement:

```
cleanedDocuments(1:10)
```

5. Lemmatize the words using the following statements:

```
cleanedDocuments = normalizeWords(cleanedDocuments,'Style','lemma');
```

- Now, again display the first ten documents using the following statement:

```
cleanedDocuments(1:10)
```

6. Remove punctuation using the using the following statement:

```
cleanedDocuments = erasePunctuation(cleanedDocuments);
```

- Now, again display the first ten documents using the following statement:

```
cleanedDocuments(1:10)
```

7. Creat bag-of-words model using the following statement:

```
bag=bagOfWords(cleanedDocuments);
```

The bag-of-words model is a simplifying representation used in natural language processing and information retrieval. In this model, a document is represented as the bag of its words, disregarding grammar and even word order but keeping multiplicity.
(see slides 47 and 48 from Lecture 7)
*(from the workspace, open and see the content of variable **bag**)*

Built in Data Sets for Text Analytics in MATLAB

https://ch.mathworks.com/help/textanalytics/ug/data-sets-for-text-analytics.html

Do yourself!
Process the Shakespeare's Sonnets (the second data set on the link given above)

Suggested Readings:

To know more about the classification system, visit the page

https://uk.mathworks.com/help/textanalytics/ug/create-simple-text-model-for-classification.html

To visualize the text data using word clouds, visit the page

https://uk.mathworks.com/help/textanalytics/ug/visualize-text-data-using-word-clouds.html

What is the purpose of text processing? To compare raw data and cleaned data, visit the page

https://uk.mathworks.com/help/textanalytics/ug/prepare-text-data-for-analysis.html